From Data to Retention:
Customer Churn Analysis & Prediction

A complete end-to-end data science solution combining data storytelling, Power BI business intelligence, and Python machine learning to understand, predict, and prevent customer churn in the telecom industry.

Power BI | Python | Scikit-learn | Random Forest | EDA | Machine Learning | Predictive Analytics

7,043	26.54%	2	8
Customers Analyzed	Churn Rate	Dashboards Built	Total Pages

Welcome to my Customer Churn Analysis Project. This project combines business intelligence and machine learning to deliver a complete churn management solution for the telecom industry. Rather than just reporting what happened, this project goes one step further by predicting who will churn next and enabling the business to act before it is too late.

Dashboard 1 — Business Analysis

Uncovers why customers are leaving using exploratory data analysis and visual storytelling across 5 pages.

Dashboard 2 — ML Predictions

Predicts who will leave next using a Random Forest classifier trained on historical telecom data across 3 pages.

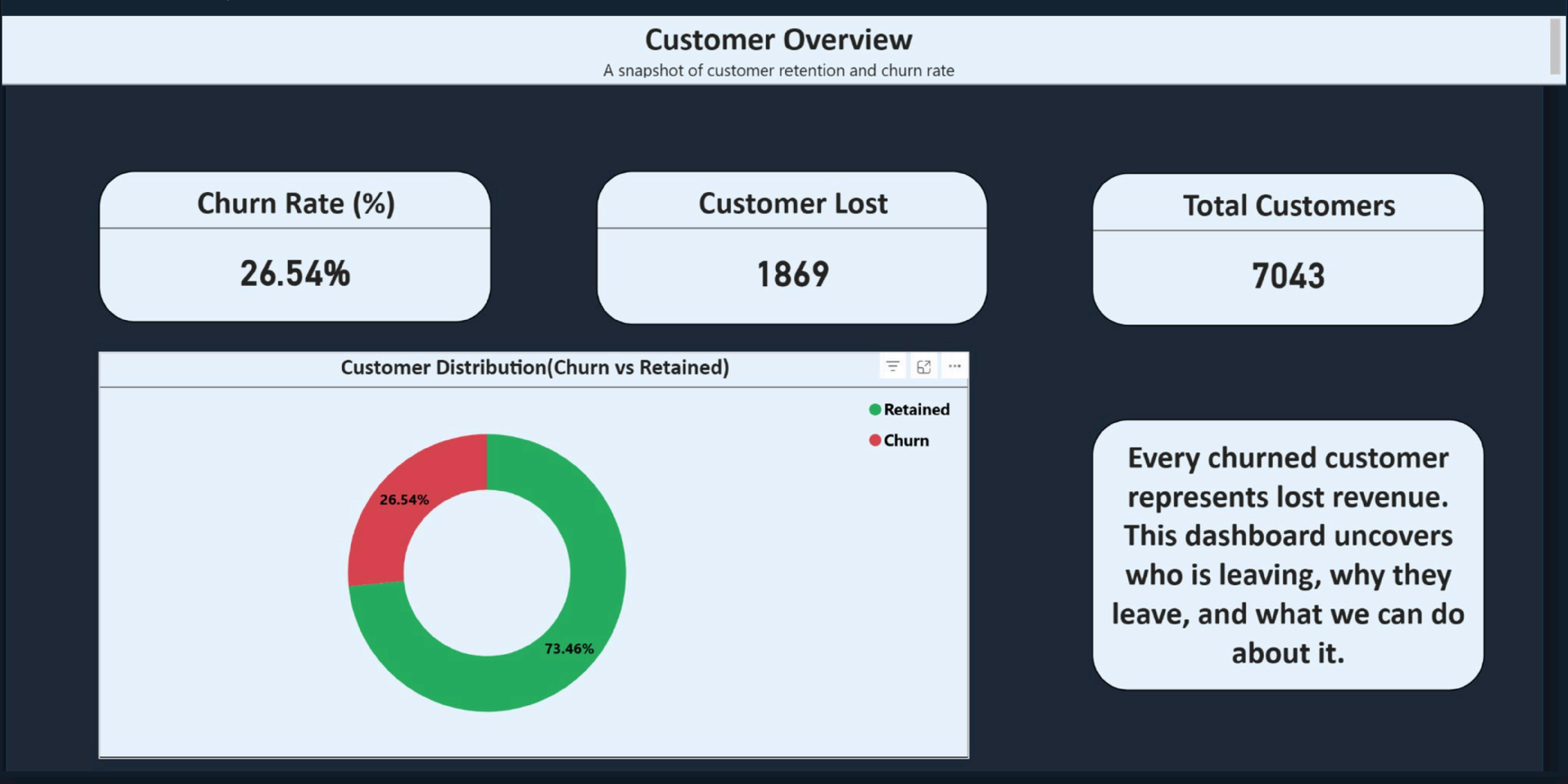
Business Analysis

Understanding the Problem — 5 Pages

PAGE 01

Customer Overview

• Customer Overview — A snapshot of customer retention and churn rate



This page provides a high level snapshot of the customer base and sets the stage for the entire analysis.

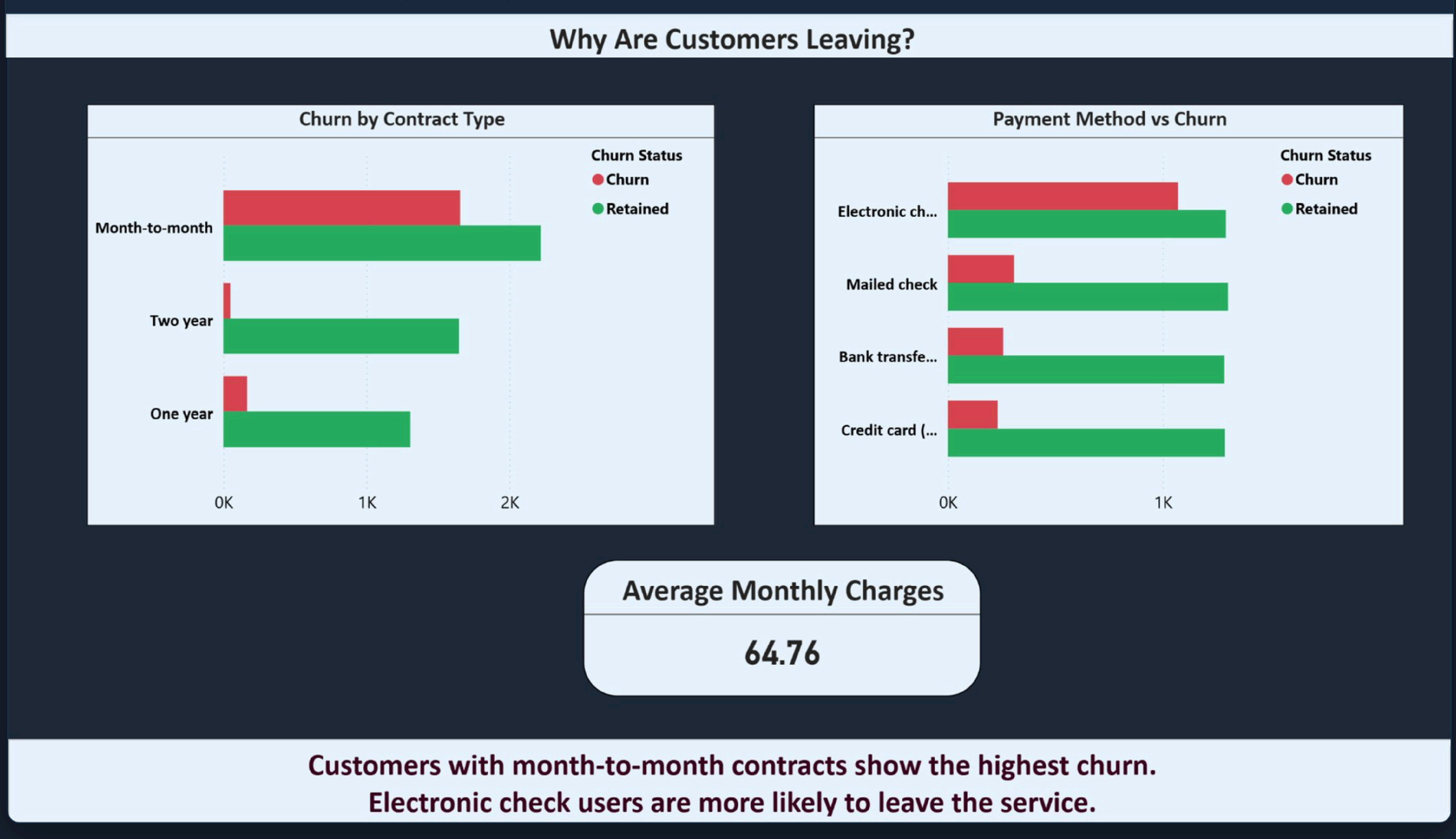
- 7,043 total customers analyzed
- 1,869 customers have already churned
- Overall churn rate stands at 26.54%
- Nearly 1 in 4 customers has left the business

> *A 26% churn rate is a significant business problem and this dashboard was built to understand it and solve it.*

PAGE 02

Why Are Customers Leaving?

• Churn by Contract Type · Payment Method vs Churn · Average Monthly Charges



Here we explore what is driving customers to leave. Two factors stand out clearly.

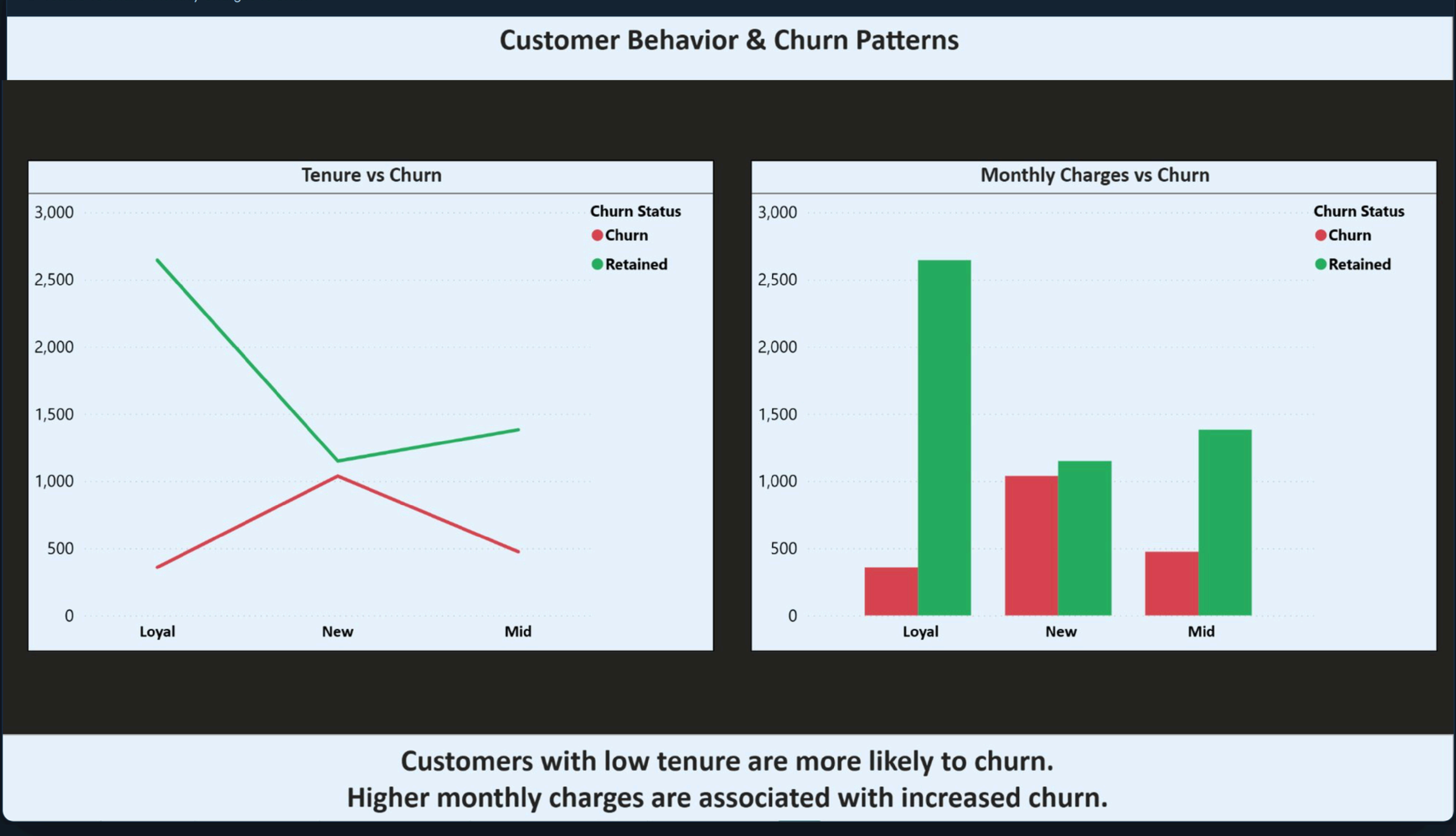
- Month to month contract holders show the highest churn rate
- Electronic check users are more likely to leave than any other payment method
- Long term contract customers show significantly lower churn
- Average monthly charges stand at 64.76

> *Contract type and payment method are strong early warning signals that the business can act on immediately.*

PAGE 03

Customer Behavior and Churn Patterns

• Tenure vs Churn · Monthly Charges vs Churn



This page examines how customer behavior and spending patterns relate to churn.

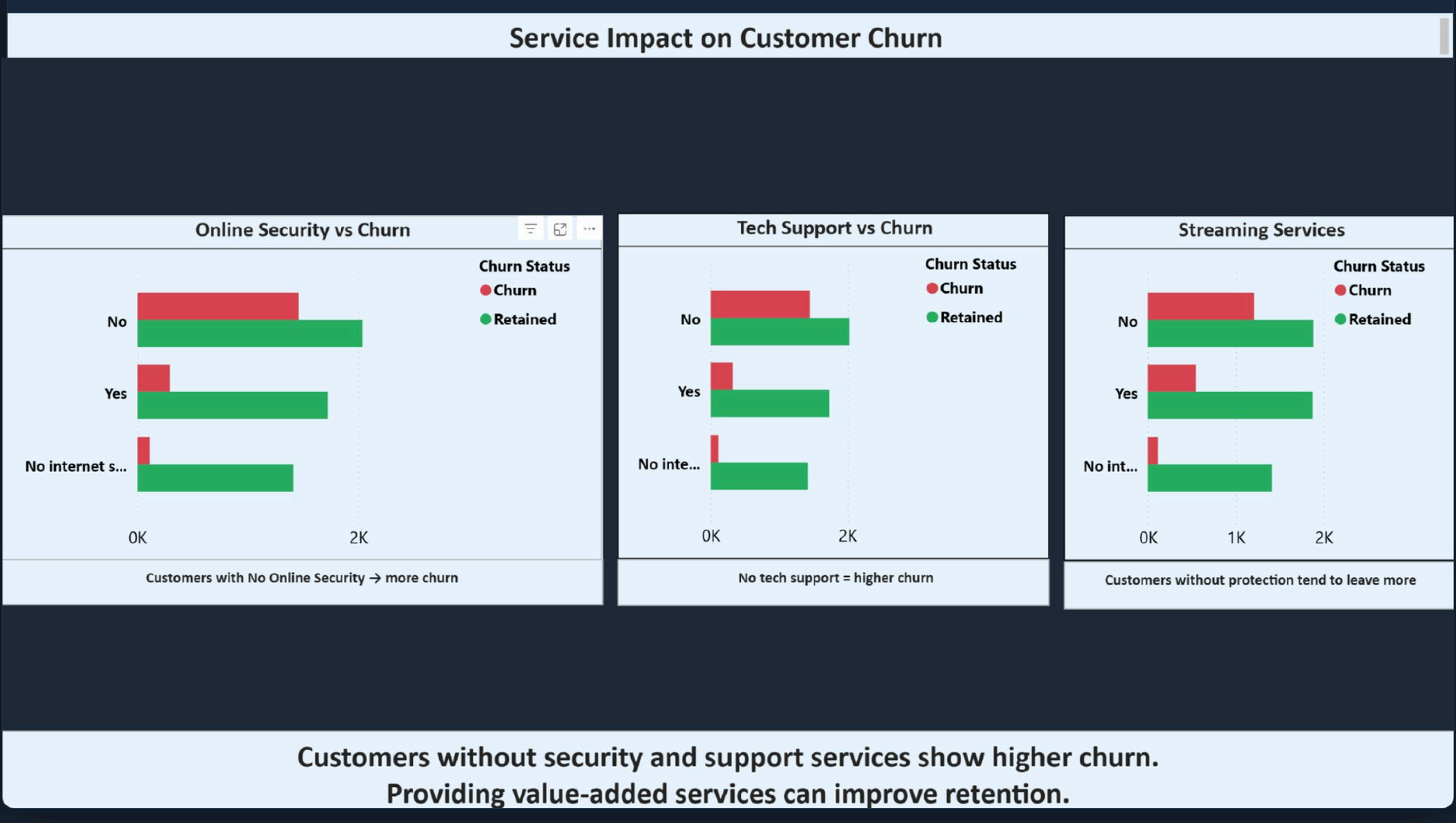
- New customers show the highest churn rate across all tenure groups
- Churn drops significantly as customer tenure increases from New to Loyal
- Mid tenure customers also show elevated churn compared to Loyal customers
- Higher monthly charges are consistently associated with increased churn

> *The first 12 months is the most critical retention window and early engagement can make or break a customer relationship.*

PAGE 04

Service Impact on Customer Churn

• Online Security vs Churn · Tech Support vs Churn · Streaming Services vs Churn



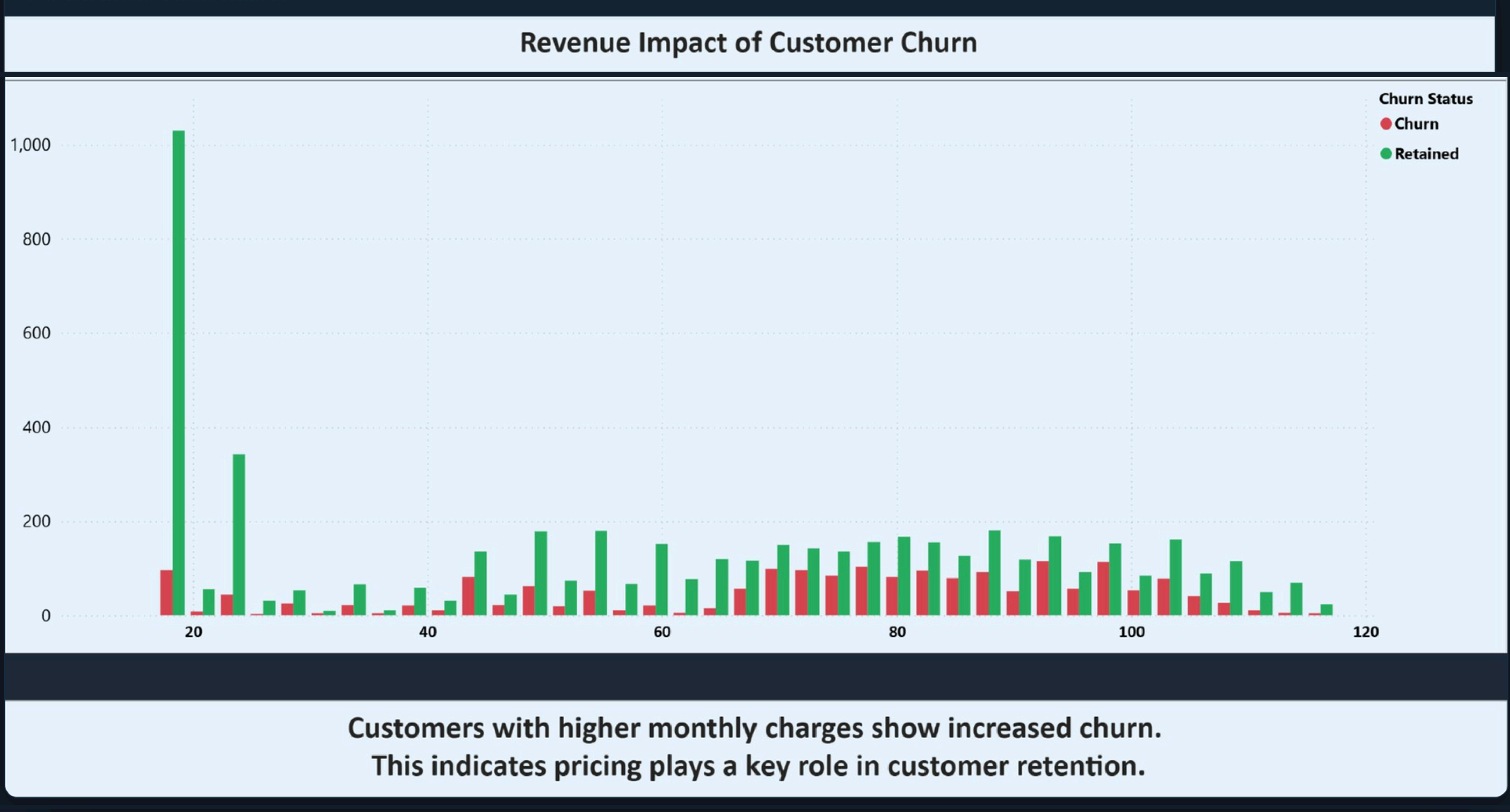
Value added services play a much bigger role in retention than most businesses realize.

- Customers without online security churn at a noticeably higher rate
- Lack of tech support is strongly linked to increased churn
- Customers without streaming services also show higher churn tendencies
- Customers with no internet service show the lowest churn across all three categories

> *These services are not just extras. They are loyalty anchors and getting customers to adopt them early is one of the most effective retention strategies available.*

Revenue Impact of Customer Churn

• Revenue Lost from Churned Customers



The final page of Dashboard 1 puts a financial lens on the churn problem.

- Revenue loss from churned customers is visible across all monthly charge ranges
- Churn is spread across both low and high charge segments
- Higher paying customers churning creates a disproportionate revenue loss
- Pricing clearly plays a key role in whether customers stay or leave

> *This page makes the business case for investing in retention. The cost of losing a high paying customer far outweighs the cost of keeping them.*

DASHBOARD 2

ML Predictions

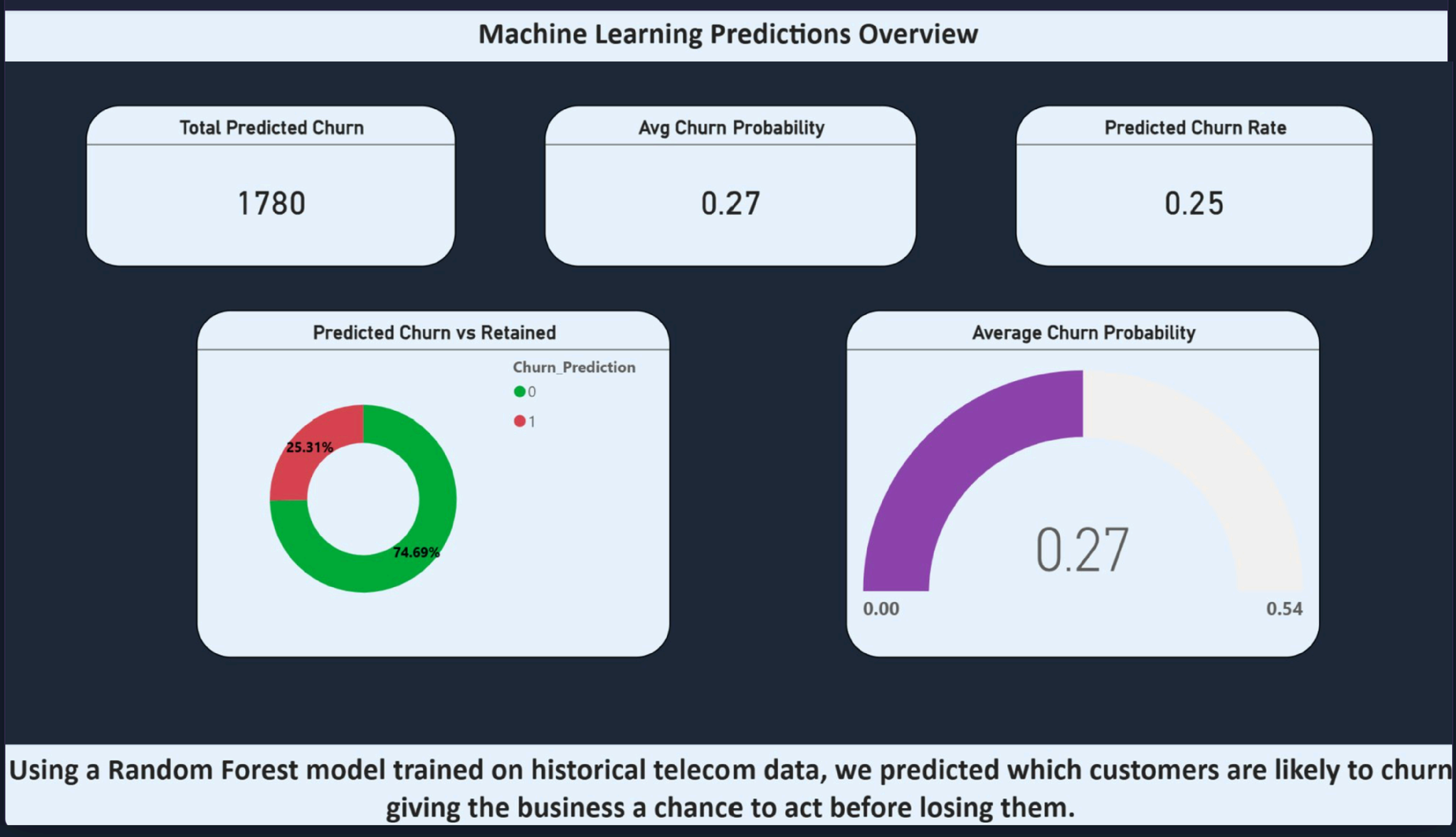
Predicting and Preventing Churn — 3 Pages

Having understood the problem in Dashboard 1 we now use machine learning to predict it. A Random Forest classifier was trained on the same telecom dataset and the predictions are brought to life in this dashboard.

ML PAGE 01

Machine Learning Predictions Overview

• Total Predicted Churn · Avg Churn Probability · Predicted Churn Rate · Donut Chart · Gauge



This page gives an immediate sense of the overall churn risk across the customer base.

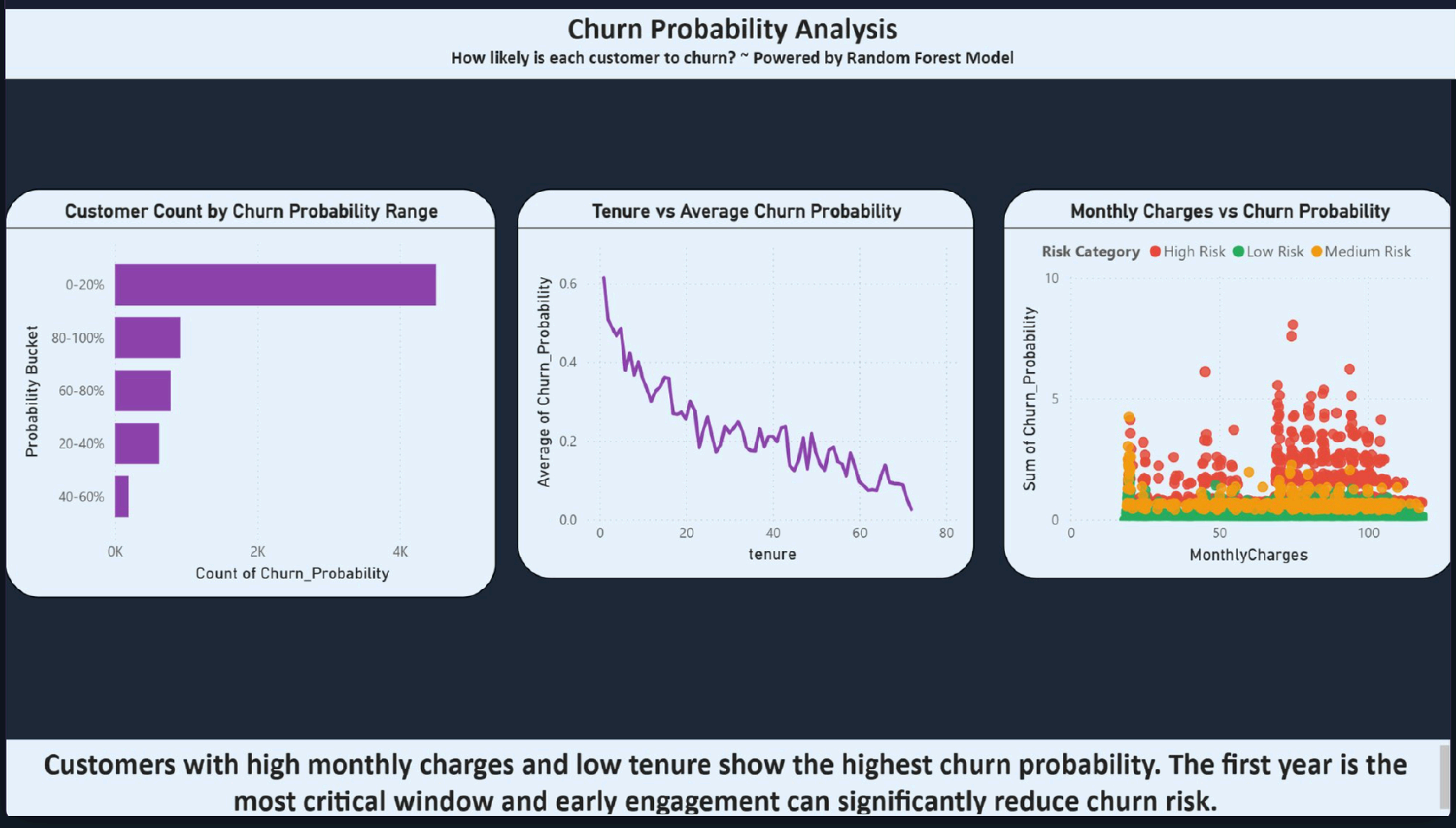
- 1,780 customers are predicted to churn by the model
- Average churn probability across all customers stands at 0.27
- Predicted churn rate is 25% which closely matches the actual 26.54%
- The gauge chart confirms the model is operating in a moderate risk zone

The model predictions closely align with actual churn confirming the Random Forest classifier is performing reliably on this dataset.

ML PAGE 02

Churn Probability Analysis

• Customer Count by Probability Range · Tenure vs Churn Probability · Monthly Charges vs Churn Probability



This page breaks down how churn probability is distributed across all customers and what drives it.

- The majority of customers fall in the 0 to 20% probability bucket confirming most are low risk
- A significant segment sits in the 80 to 100% bucket and these customers need urgent attention
- Churn probability drops steadily and consistently as customer tenure increases
- Higher monthly charges push churn probability up across all risk categories

Machine learning confirms and quantifies exactly what Dashboard 1 discovered through visual analysis.

ML PAGE 03

Risk Segmentation

• High Risk · Medium Risk · Low Risk · Risk by Contract · Risk by Payment · Top 20 Customers Table

tenure	Contract	MonthlyCharges	Risk Category	Churn_Probability
61	2	106.60	High Risk	0.75
69	2	106.50	High Risk	0.74
71	2	106.00	High Risk	0.74
72	2	54.20	High Risk	0.74
5	1	70.00	High Risk	0.75
12	1	95.50	High Risk	0.76
13	1	75.30	High Risk	0.75
33	1	93.80	High Risk	0.82
33	1	95.80	High Risk	0.75
54	1	82.45	High Risk	0.74
54	1	105.40	High Risk	0.75
56	1	111.95	High Risk	0.75
58	1	104.65	High Risk	0.82
58	1	108.85	High Risk	0.82
59	1	110.15	High Risk	0.74
62	1	96.75	High Risk	0.75
64	1	105.65	High Risk	0.73
66	1	104.05	High Risk	0.74
66	1	110.85	High Risk	0.75
67	1	87.40	High Risk	0.75
70	1	109.50	High Risk	0.73
Total				

The final page translates model predictions into direct business action.

- 1,442 customers are classified as High Risk
- 461 customers fall in the Medium Risk category
- 5,129 customers are in the Low Risk zone
- Contract type 0 (month to month) dominates the High Risk segment
- The top 20 table identifies exact customers with churn probability above 0.73

This page bridges the gap between data science and business strategy and turns a prediction into a clear action plan.

FINAL TAKEAWAYS

Conclusion and Recommendations

This project demonstrates how combining Power BI dashboards with a Python machine learning model can move a business from reactive reporting to proactive prediction.

THREE KEY CHURN DRIVERS

- Short term month to month contracts
- High monthly charges
- Low customer tenure

RECOMMENDED ACTIONS

- Contact the 1,442 High Risk customers immediately with personalized retention offers
- Incentivize month to month customers to upgrade to longer contracts
- Focus retention efforts within the first 12 months of a customer relationship
- Offer better value to high paying customers showing early churn signals

Dashboard 1 tells us what is happening and why. **Dashboard 2** tells us who will churn next and what to do about it. Together they form a complete end to end churn management solution that is both analytically rigorous and business ready.